

Correlation and Linear Regression

180 minutes · 6 self-check questions

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Modelling a continuous outcome: what a regression coefficient means, how to handle categorical predictors, and how to check the model is entitled to the conclusions you want from it.

Learning objectives

- Distinguish correlation from regression and know when each is appropriate
- Interpret a slope, an intercept and R-squared in the units of the problem
- Include categorical predictors correctly using dummy coding
- Extend to multiple regression and interpret adjusted coefficients
- Check the assumptions using residual plots
- Report a regression model completely

Correlation

- Pearson r measures the strength and direction of a linear relationship between two continuous variables
- It runs from -1 to $+1$; 0 means no linear relationship, not no relationship at all
- Spearman ρ is the rank-based alternative for monotonic but non-linear relationships, or with outliers
- r -squared is the proportion of variance in one variable explained by the other
- Always plot the scatterplot before quoting r — very different patterns give identical coefficients
- Correlation is symmetric and makes no prediction; regression is directional and does

Simple linear regression

- The model is $y = b_0 + b_1 \cdot x + \text{error}$
- b_1 is the estimated change in y for a one-unit increase in x
- b_0 is the predicted y when x is zero — often meaningless, as with a predicted birth weight at age zero
- Centring a predictor (subtracting its mean) makes the intercept interpretable as the value at the average
- R-squared is the proportion of variance explained; a low value does not mean the model is wrong, only that much is unexplained
- In R: `fit <- lm(bp ~ age, data = dat); summary(fit); confint(fit)`
- Always report the coefficient with its confidence interval, not just its p-value

Categorical predictors

- A binary predictor enters as 0/1, and its coefficient is the difference between the two groups
- A k-category predictor needs k-1 dummy variables, with one level as the reference
- Every coefficient is then interpreted as the difference from that reference level
- Choose a reference that makes sense — usually the largest group, or the untreated one
- In R declare it a factor and `lm()` handles the coding: `dat$edu <- factor(dat$edu)`
- Entering an unordered categorical variable as a plain number is a common and serious error: it forces a straight-line effect across arbitrary category codes

IN PRACTICE

Coding education as 1 to 5 and putting it in as numeric assumes the step from none to primary equals the step from SHS to tertiary, and that both are one unit of the same thing. Almost never true — use factors.

Multiple regression

- Each coefficient is the effect of that predictor holding the others constant
- "Adjusted for" and "holding constant" mean the same thing here
- Adding a confounder changes the coefficient of interest; that change is the confounding
- Adjusted R-squared penalises adding predictors that do not help
- Do not select variables purely by p-value; stepwise selection produces unstable models and biased intervals
- Choose predictors from subject knowledge and the conceptual framework built in Part 1
- A rough guide: at least 10-15 observations per predictor, more for binary outcomes

READING A REGRESSION TABLE

Predictor	b (95% CI)	p	Interpretation
Age (years)	0.42 (0.28 to 0.56)	<0.001	Each extra year adds 0.42 mmHg
Sex (female vs male)	-3.10 (-5.9 to -0.3)	0.030	Women 3.1 mmHg lower than men
BMI (kg/m ²)	0.88 (0.55 to 1.21)	<0.001	Each BMI unit adds 0.88 mmHg
Education: tertiary vs none	-2.40 (-5.1 to 0.3)	0.081	Lower, but interval includes no difference

Checking the model

- Linearity: residuals versus fitted values should show no pattern; a curve means the relationship is not linear
- Constant variance: the same plot should not fan out — a funnel shape is heteroscedasticity
- Normality of residuals: a Q-Q plot should follow the diagonal. It is the residuals that matter, not the raw data
- Independence: cannot be checked from residuals alone; it follows from the design
- Influential points: Cook's distance identifies observations that move the coefficients substantially
- Multicollinearity: a variance inflation factor above about 5 to 10 means predictors are too closely related to separate
- In R: `plot(fit)` gives the four diagnostic plots; `vif(fit)` from the `car` package gives VIFs

Worked examples

Reading a regression table out loud

A model of systolic blood pressure gives: age 0.42 (0.28 to 0.56); female -3.10 (-5.9 to -0.3); BMI 0.88 (0.55 to 1.21); tertiary education -2.40 (-5.1 to 0.3).

1. Age: each additional year is associated with 0.42 mmHg higher SBP, holding sex, BMI and education constant.
2. Female: women average 3.1 mmHg lower than men, adjusted for the others. The reference category is male.
3. BMI: each additional kg/m² adds 0.88 mmHg — over 5 units, about 4.4 mmHg.
4. Tertiary education: 2.4 mmHg lower than the reference level, but the interval crosses zero, so this is compatible with no difference.
5. Every statement needs "holding the others constant" — that is what adjustment means.
6. Reporting only "age was significant" throws away the magnitude, which is the part a clinician can use.

THE POINT

A coefficient is a sentence in the units of the problem, not a p-value. If you cannot say it out loud in those units, you cannot yet interpret your own model.

The education variable entered as a number

Education has five categories coded 1 to 5. It is entered into the regression as numeric and returns a neat coefficient of -1.2 per level.

1. That single coefficient assumes each step has the same effect: none→primary equals SHS→tertiary.
2. It also assumes the categories are equally spaced on some underlying scale, which nobody has established.
3. Declaring it a factor — `dat$edu <- factor(dat$edu)` — estimates four coefficients against a reference.
4. Refitted, the effect turns out to be flat across the first three levels and then drops sharply at tertiary.
5. The linear version averaged that into a misleading constant slope and understated the tertiary effect.
6. If a single-number summary is genuinely wanted, test for trend explicitly and say that is what you did.

THE POINT

Numeric coding of an unordered or unequally spaced categorical variable imposes a straight line nobody checked. Use factors, and let the data show the shape.

Key terms

Pearson r

A measure of linear association between two continuous variables, from -1 to +1.

Regression coefficient

The estimated change in the outcome per one-unit increase in a predictor.

Dummy variable

A binary indicator representing one level of a categorical predictor against a reference.

Adjusted

Estimated while holding the other predictors in the model constant.

Heteroscedasticity

Non-constant variance of residuals across fitted values.

Multicollinearity

Predictors so correlated that their separate effects cannot be estimated reliably.

Cook's distance

A measure of how much an observation influences the fitted coefficients.

Activity

Fit a regression with one continuous and one categorical predictor on your own data or a supplied dataset. Write one sentence interpreting each coefficient in the units of the problem, then run the four diagnostic plots and say which assumption you are least comfortable with.

Further reading

- **Vittinghoff et al. — Regression Methods in Biostatistics** — The standard applied text; strong on confounding and model building.
- **Altman & Bland — "Correlation, regression and repeated data"** — BMJ Statistics Notes, short and corrective.
- **Harrell — Regression Modeling Strategies** — Advanced; definitive on why stepwise selection should be avoided.

Self-check questions

1. In a regression of blood pressure on age, the coefficient for age is 0.42. This means:
 - a. 42% of blood pressure is explained by age
 - b. Each additional year of age is associated with a 0.42 mmHg higher blood pressure
 - c. Age and blood pressure correlate at $r = 0.42$
 - d. Blood pressure rises 0.42% per year
2. Education has five ordered categories. Entering it as a numeric variable 1-5 assumes:
 - a. Nothing problematic
 - b. That each step between adjacent categories has the same effect on the outcome
 - c. That education is normally distributed
 - d. That the reference category is "none"
3. A residuals-versus-fitted plot shows a clear funnel widening to the right. This indicates:
 - a. Non-normal residuals
 - b. Heteroscedasticity — the variance is not constant across fitted values
 - c. Multicollinearity
 - d. An omitted interaction
4. Adding a confounder to a model changes the coefficient of interest from 5.2 to 2.1. This means:
 - a. The model is misspecified

- b. Much of the crude association was explained by that confounder
 - c. The confounder should be removed
 - d. The sample is too small
 - 5. A Q-Q plot in regression diagnostics assesses the normality of:
 - a. The outcome variable
 - b. The residuals
 - c. The predictors
 - d. The fitted values
 - 6. Stepwise selection by p-value is discouraged mainly because it:
 - a. Is computationally slow
 - b. Produces unstable models with biased coefficients and confidence intervals that are too narrow
 - c. Cannot handle categorical predictors
 - d. Requires very large samples
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Answers

- 1. **b.** A regression coefficient is the change in the outcome per one-unit change in the predictor, in the outcome's own units. Variance explained is R-squared; correlation is r.
 - 2. **b.** Numeric entry imposes a constant linear effect per category step. Declaring it a factor estimates a separate coefficient for each level against a reference, which is almost always the honest choice.
 - 3. **b.** A funnel means spread changes with the fitted value. Coefficients stay unbiased but standard errors are wrong; a transformation or robust standard errors is the usual fix.
 - 4. **b.** That change is exactly what confounding looks like. The adjusted estimate is the effect holding the confounder constant, and reporting both crude and adjusted lets the reader see how much it mattered.
 - 5. **b.** Linear regression assumes normally distributed errors, not normally distributed raw data. A skewed outcome can still yield well-behaved residuals once predictors are in the model.
 - 6. **b.** Letting the data choose the model and then reporting intervals as if it had been pre-specified understates uncertainty, and small changes in the data can produce a different model. Choose predictors from subject knowledge instead.
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